Explainable AI with SHAP

Explainable AI

- Explainable AI/ML aims to create models and methods that are transparent, interpretable, and explainable
- One goal is to help ensure that AI and ML are ethical and accountable
- There are several techniques that aim to make ML models explainable
  - Interpretable models
  - Model agnostic methods (local and global)
  - A common model agnostic local method:
    - SHAP (**SH**apley **A**dditive ex**P**lanations)

Shapley Values

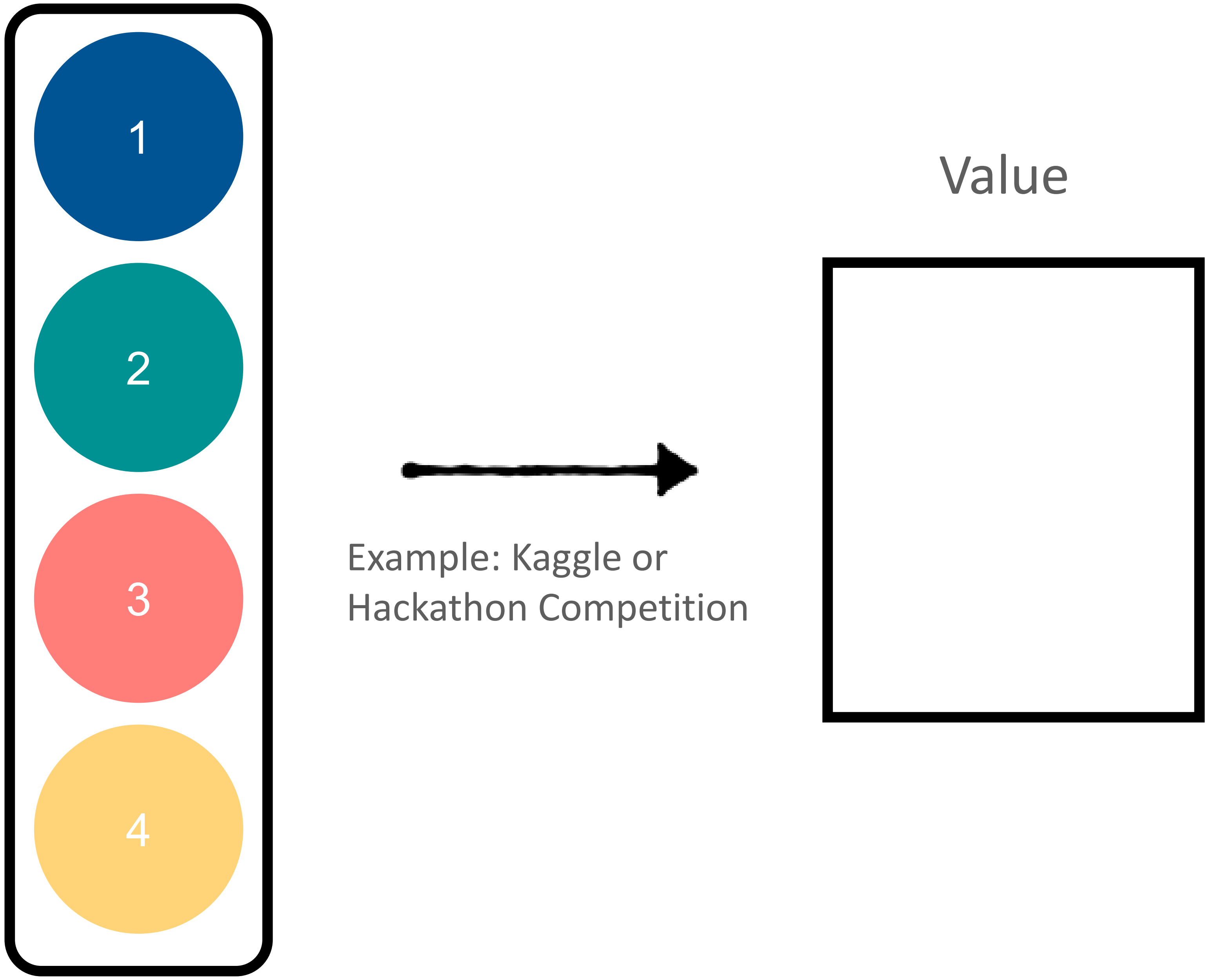
Shapley Values

- Shapley values are a concept in cooperative game theory
- Introduced by Lloyd Shapley in 1951
- Shapley won the Nobel Prize in Economics for this contribution in 2012
- Shapley values try to solve this problem:
    - *Setup*: A coalition of players cooperate to produce a value
    - How much did each player contribution to the value?
    - What payoff can each player reasonably expect?

Shapley Values

Coalition

Value

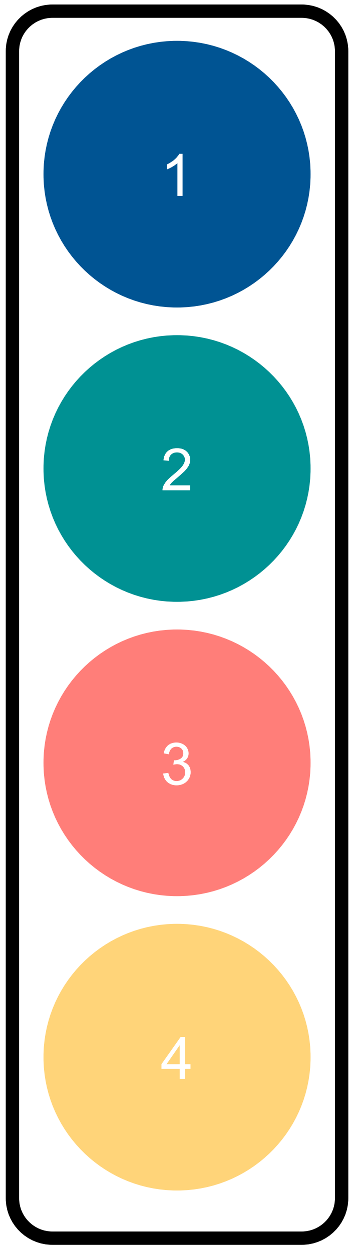


Example: Team

Example: Prize is $10,000

Shapley Values

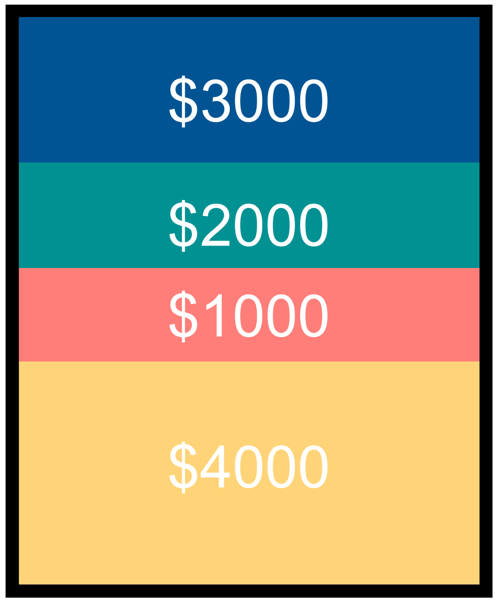
Coalition



Example: Team

Example: Kaggle or Hackathon Competition

Value



Example: Prize is $10,000

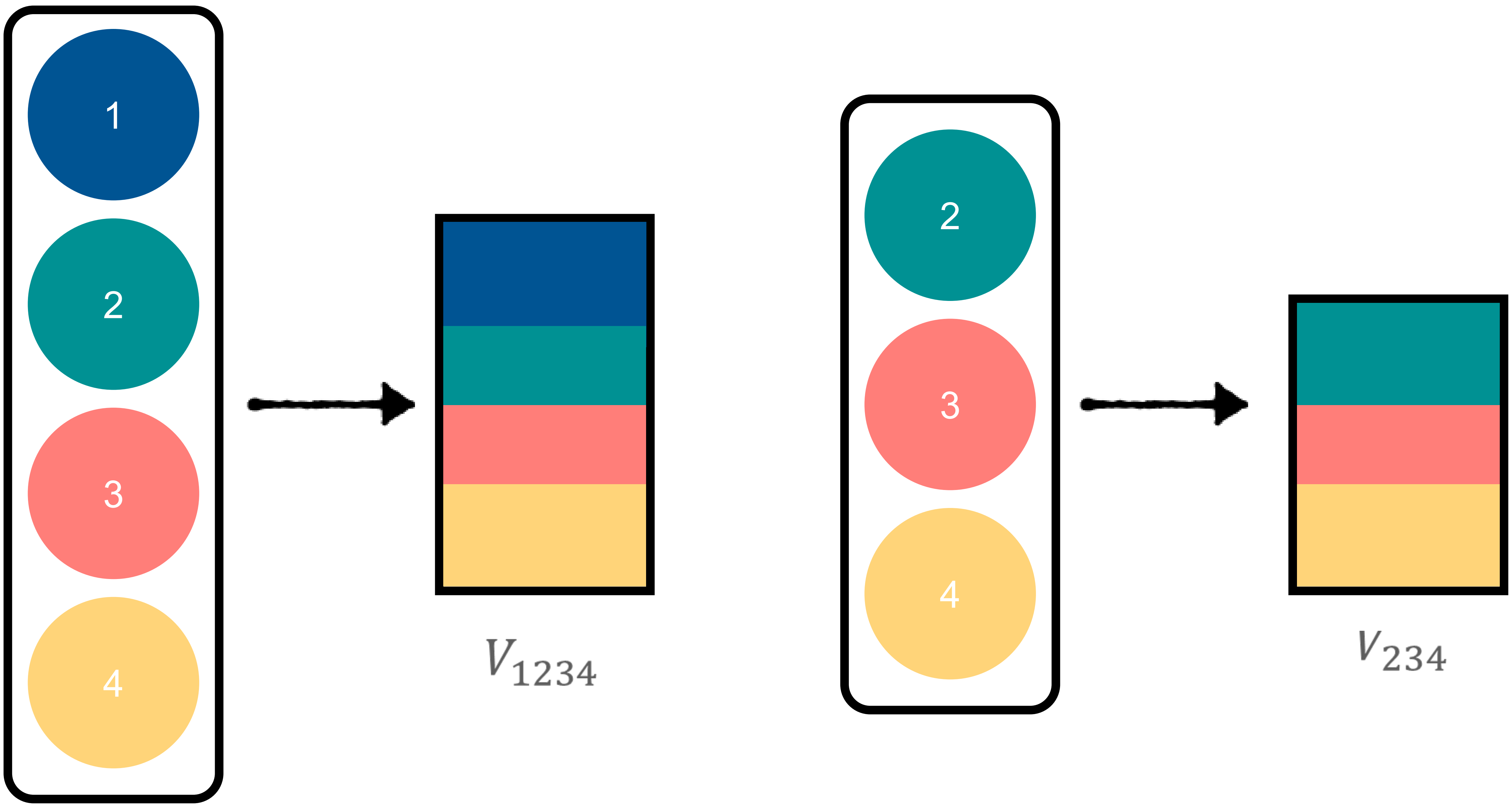
How should each player be rewarded?

What if there are interactions between team members?

(Certain combinations of members work better or worse than the sum of the parts)

Shapley Values

- For each individual, compute the marginal contribution by comparing the value for a coalition with the member and without the member



Shapley Values

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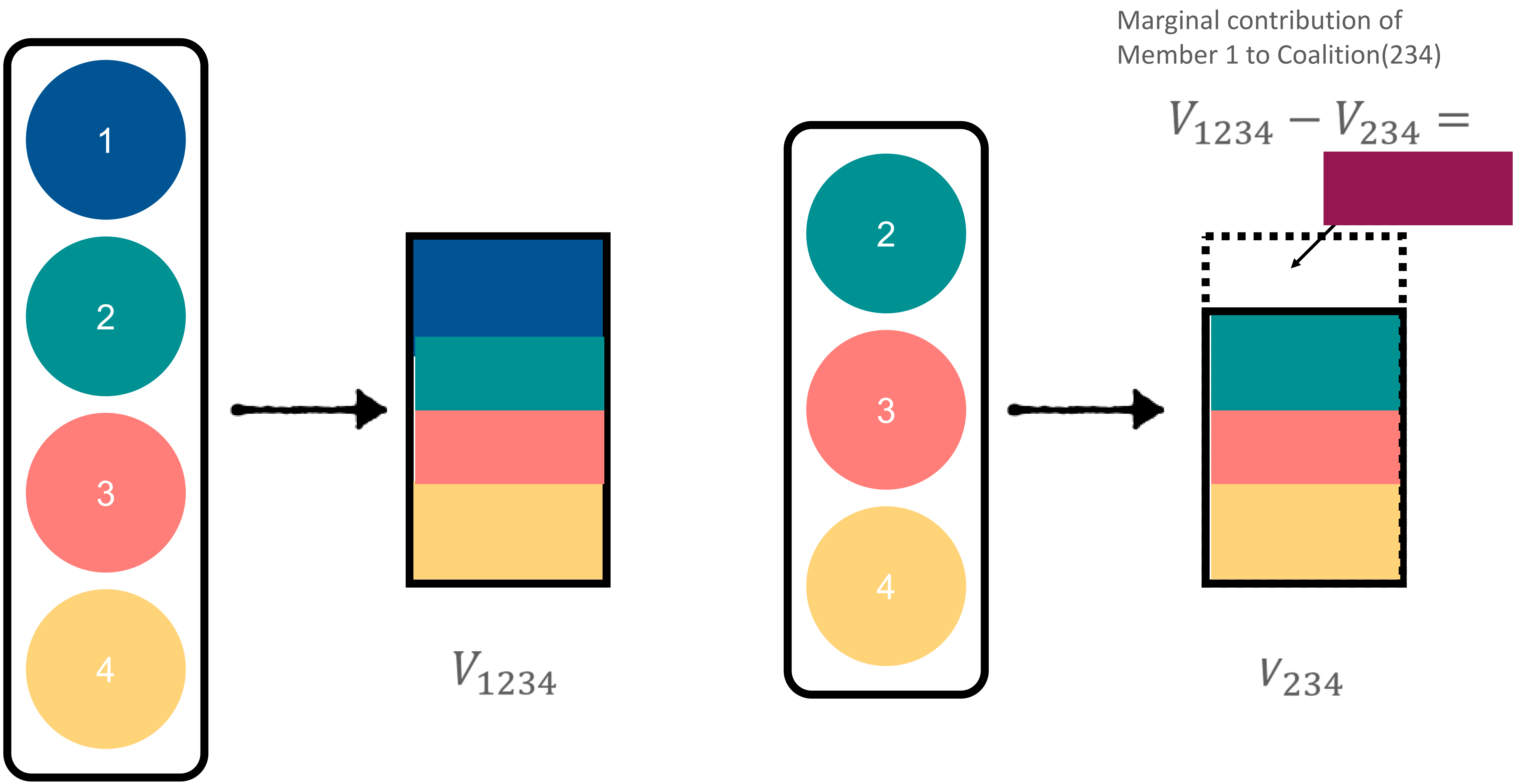
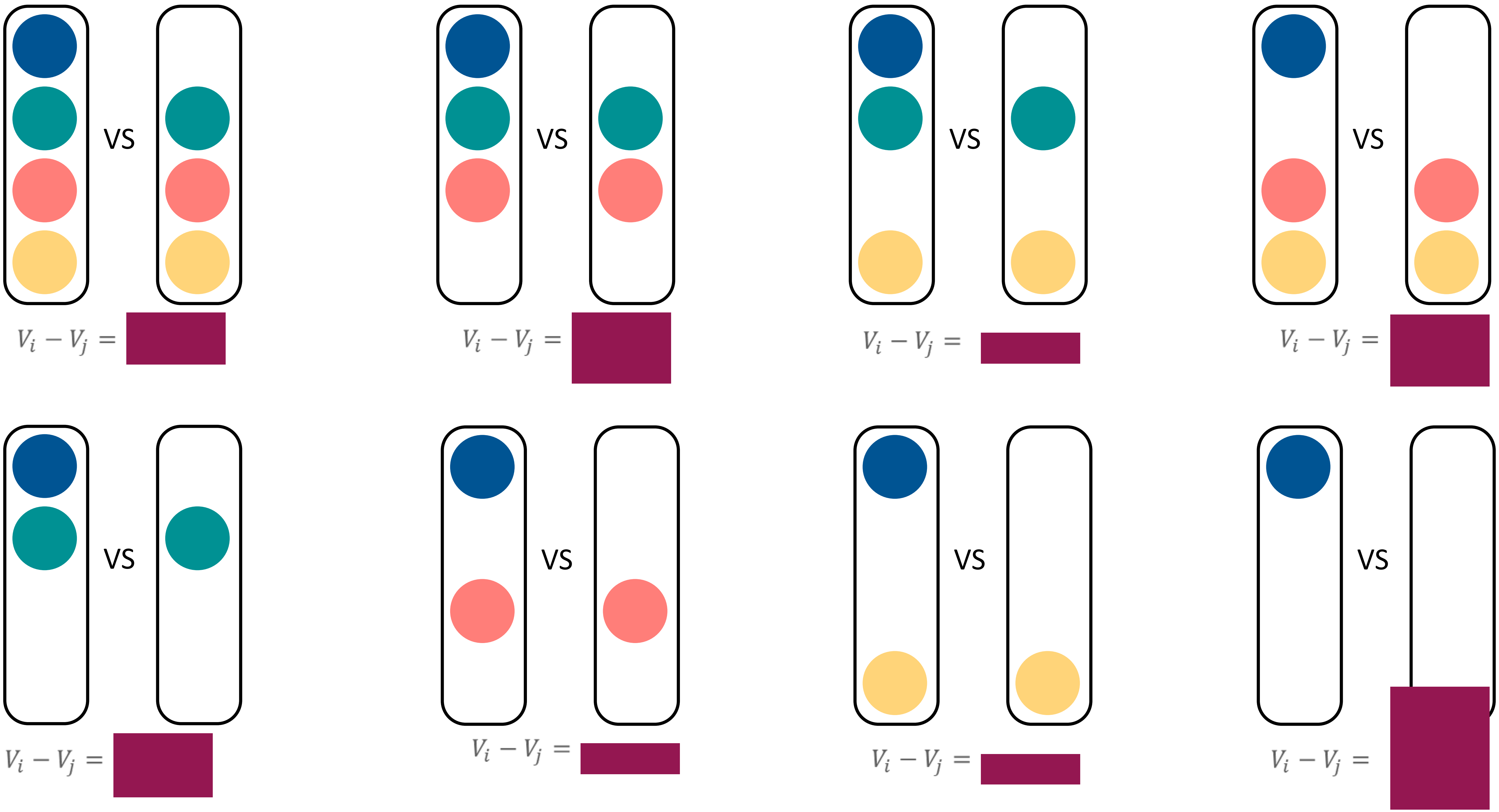


Shapley Values

- Compute the marginal contribution of each member to all combinations of coalitions with and without that member

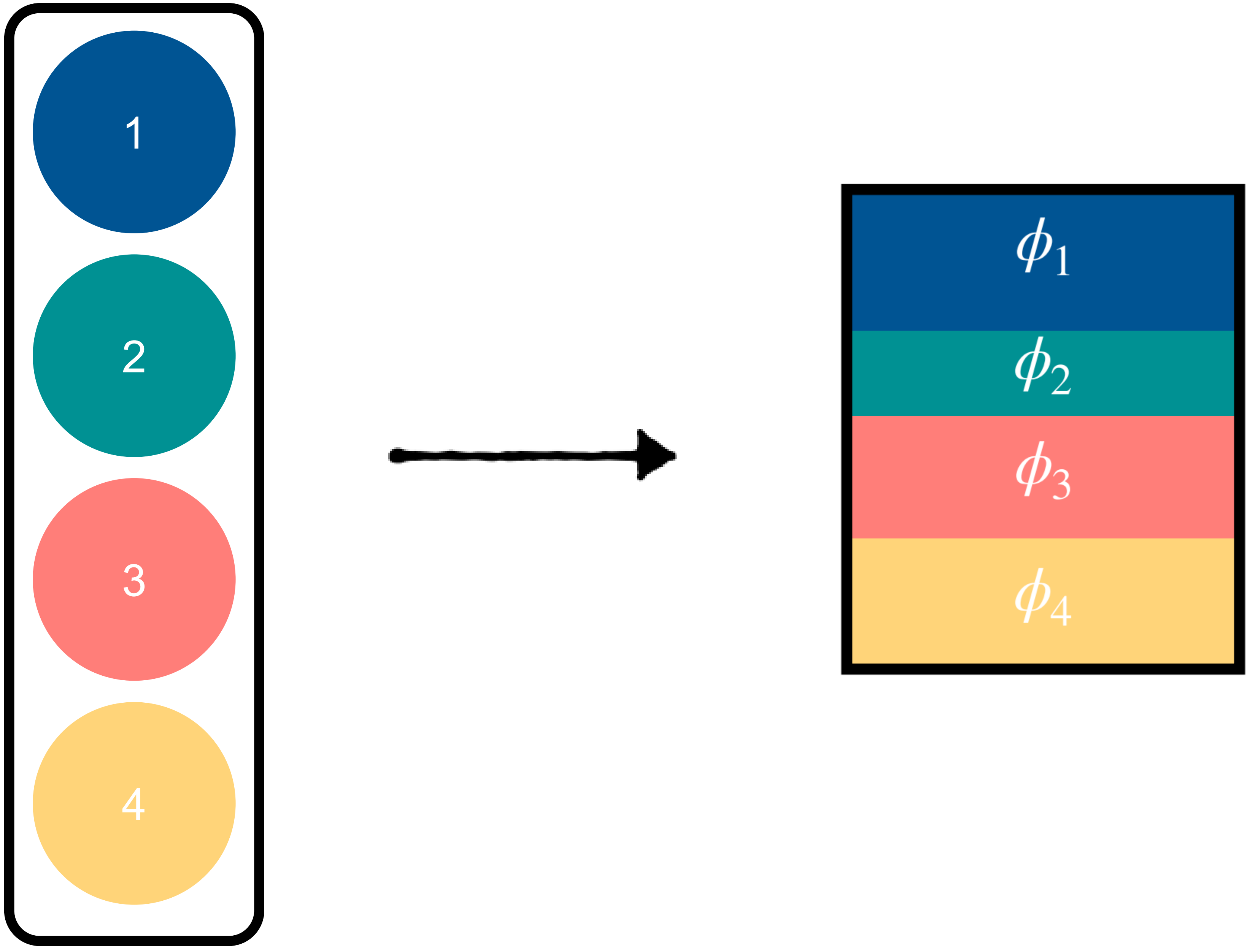


Shapley Values

- Compute the marginal contribution of each member to all combinations of coalitions with and without that member
- The Shapley value is the average marginal contribution for each member averaged across the coalition comparisons

Shapley Values

- Repeat for all coalition members to get the Shapley values for each member



In Machine Learning

- Select a feature, say x_j, to determine its importance
  - Consider all possible subsets: Compute predictions with and without x_j for every subset of remaining features
  - Compute marginal contributions: Compute the change in prediction caused by adding x_j
  - Weight the contributions based on how often they appear in permutations
  - Weighted sum of marginal contribution is the Shapley Value
- SHAP is a practical framework that builds on Shapley Values but adds optimizations and approximations

SHapley Additive exPlanations

Shapley applied to explainable AI

- In 2017, Lundberg and Lee extended the idea of Shapley values to explainable machine learning

A Unified Approach to Interpreting Model Predictions

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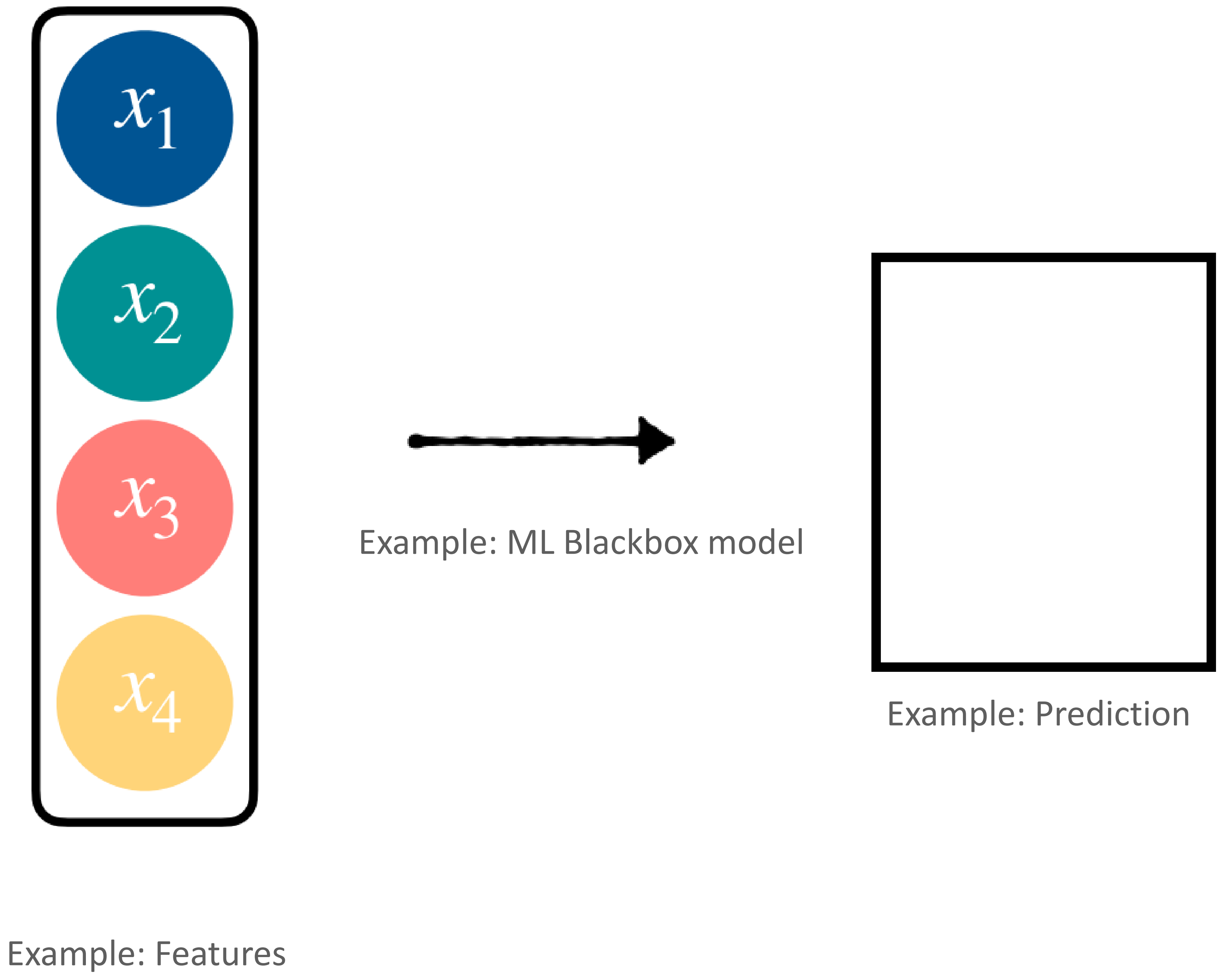
Abstract

Understanding why a model makes a certain prediction can be as crucial as the prediction's accuracy in many applications. However, the highest accuracy for large modern datasets is often achieved by complex models that even experts struggle to interpret, such as ensemble or deep learning models, creating a tension between *accuracy* and *interpretability*. In response, various methods have recently been proposed to help users interpret the predictions of complex models, but it is often unclear how these methods are related and when one method is preferable over another. To address this problem, we present a unified framework for interpreting predictions, SHAP (SHapley Additive exPlanations). SHAP assigns each feature an importance value for a particular prediction. Its novel components include: (1) the identification of a new class of additive feature importance measures, and (2) theoretical results showing there is a unique solution in this class with a set of desirable properties. The new class unifies six existing methods, notable because several recent methods in the class lack the proposed desirable properties. Based on insights from this unification, we present new methods that show improved computational performance and/or better consistency with human intuition than previous approaches.

SHAP



x_1
x_2
x_3
x_4
Example: ML Blackbox model
Example: Prediction
Example: Features

SHAP values

For a given input x, the SHAP value for feature j is:

$$\phi_j(x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} [f_x(S \cup \{j\}) - f_x(S)]$$

- F is the set of all features
- S is a subset of features excluding x_j
- $f(S)$ is the model's prediction when only features in S are used
- $f(S \cup \{j\})$ is the model's prediction when x_j is added to S
- The weighting factor ensures that larger subsets are not over represented

Properties of SHAP

- These properties are directly derived from the axioms of Shapley values in cooperative game theory
  - Efficiency (sum to model output) $\sum_{i=1}^{p} \phi_i(x) = f(x) - \mathbb{E}[f(X)]$
  - Symmetry (fairness) $\phi_i = \phi_j$ if features i, j contribute equally
  - Additivity (consistency)
- SHAP is the only feature attribution method that satisfies these properties
  - Mathematically justifies and makes SHAP more trustworthy compared to other methods

Properties of SHAP: Efficiency

- $\sum_j \phi_j = f(x) - E[f(X)]$
  - The sum of all SHAP values for a given instances equals the difference between the model's prediction and the baseline expectation
  - This ensures that all feature contributions fully explain the model's output

Properties of SHAP: Symmetry

- If $f(S \cup \{j\}) = f(S \cup \{k\})$ for all S, then $\phi_j = \phi_k$
  - If two features contribute equally to the prediction in all possible subsets, they will have equal SHAP values
  - This ensures fairness

Properties of SHAP: Additivity

- Mathematical Consistency
  - If a model changes to depend more on a feature, the SHAP value for that feature should not decrease
  - This ensure that SHAP values are consistent across different models

SHapley **Additive** exPlanations

- *(Slightly different meaning than the mathematical consistency property)*
- SHAP is an additive explanation model
  - The model's prediction is decomposed into a sum of contributions from each feature:

$$f(x) = E[f(X)] + \sum_j \phi_j$$

    - $E[f(X)]$ is the average prediction (model's average output if no features were used)
- This allows us to break down model predictions into understandable contributions

Types of SHAP

- Kernel SHAP *(slow but model agnostic)*
  - Works for any model
  - Uses random sampling and wtd regression to estimate values
- Tree SHAP *(fast but only works for tree based models)*
  - Optimized for tree models (RF, Boosting)
  - Uses dynamic programming to compute SHAP value exactly
- Deep SHAP *(faster than kernel SHAP)*
  - Optimized for neural networks
  - Uses back propagation and DeepLIFT to estimate SHAP values

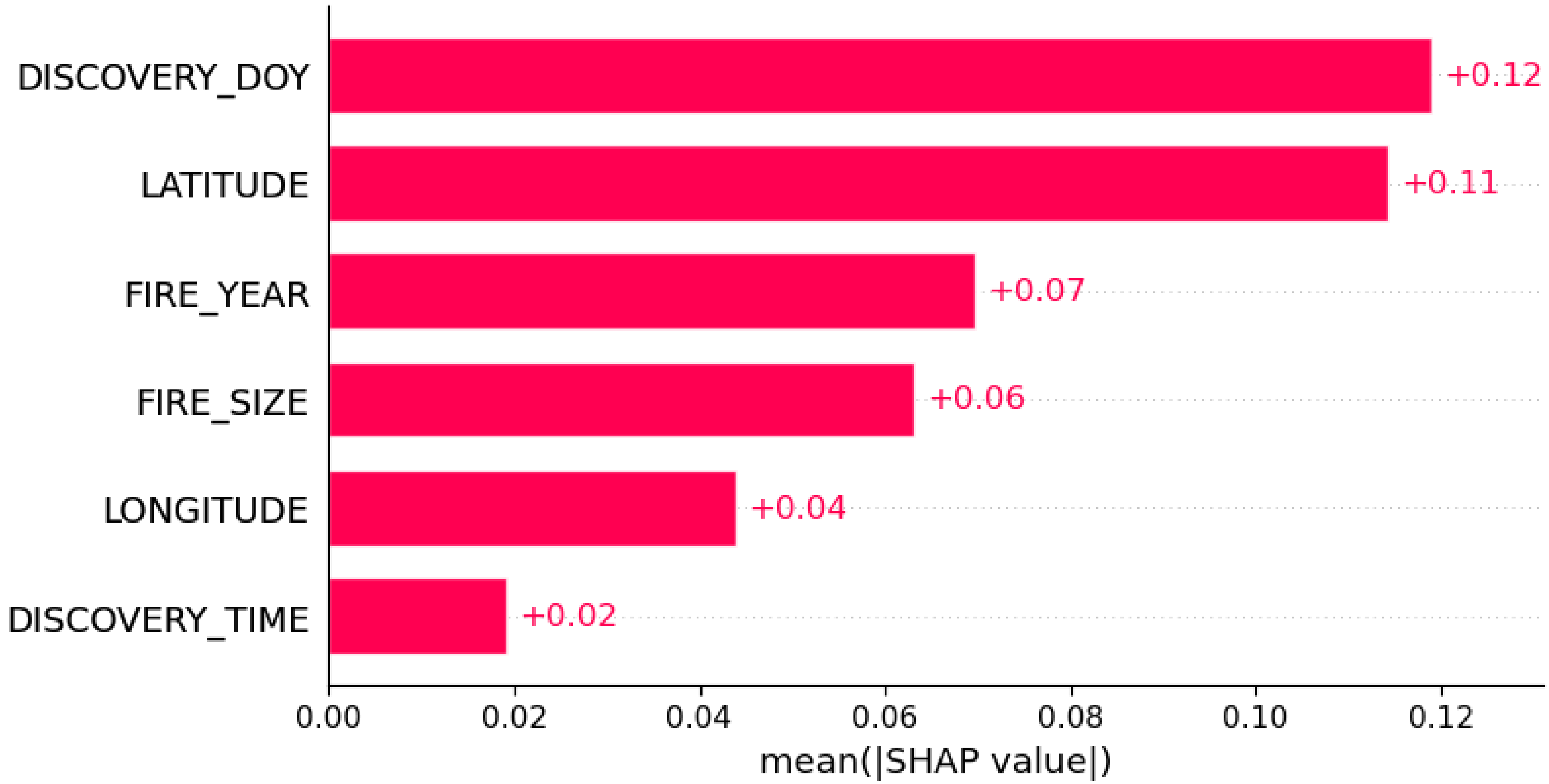
Local Explainability

- SHAP is a local explanation technique
  - Designed to explain how each feature contributes to the final prediction of *an individual prediction*
- Global explanations can be computed by aggregation

Example:
Human- vs Natural-Started Fire

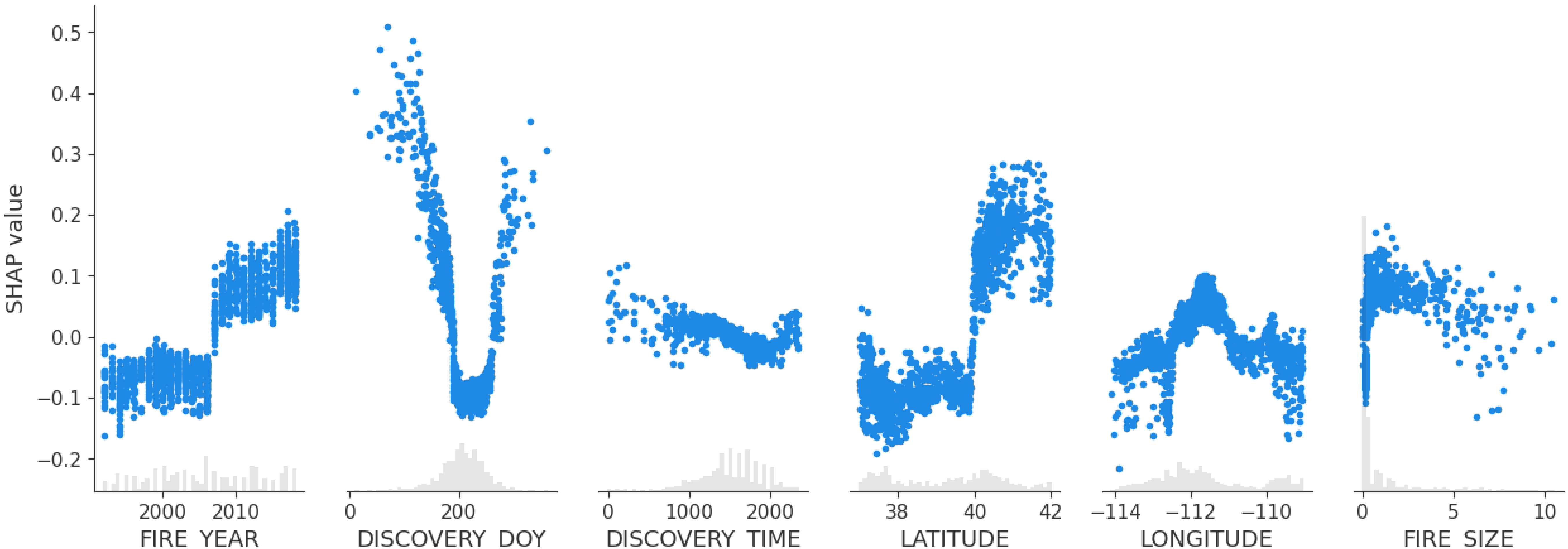
Global Importances:

Mean absolute SHAP value over all the samples



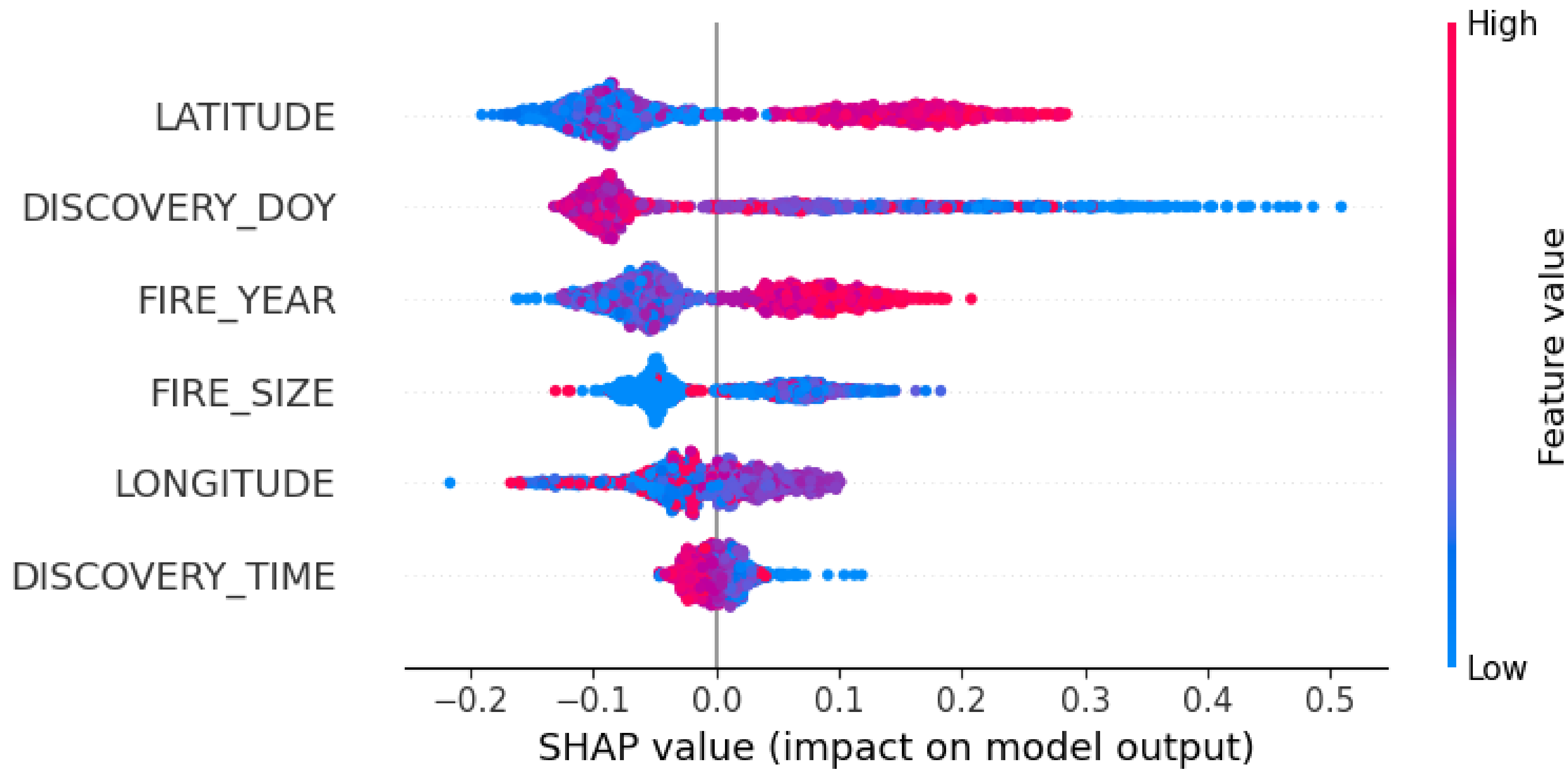
Global Importances:

Scatterplots of feature value vs SHAP value



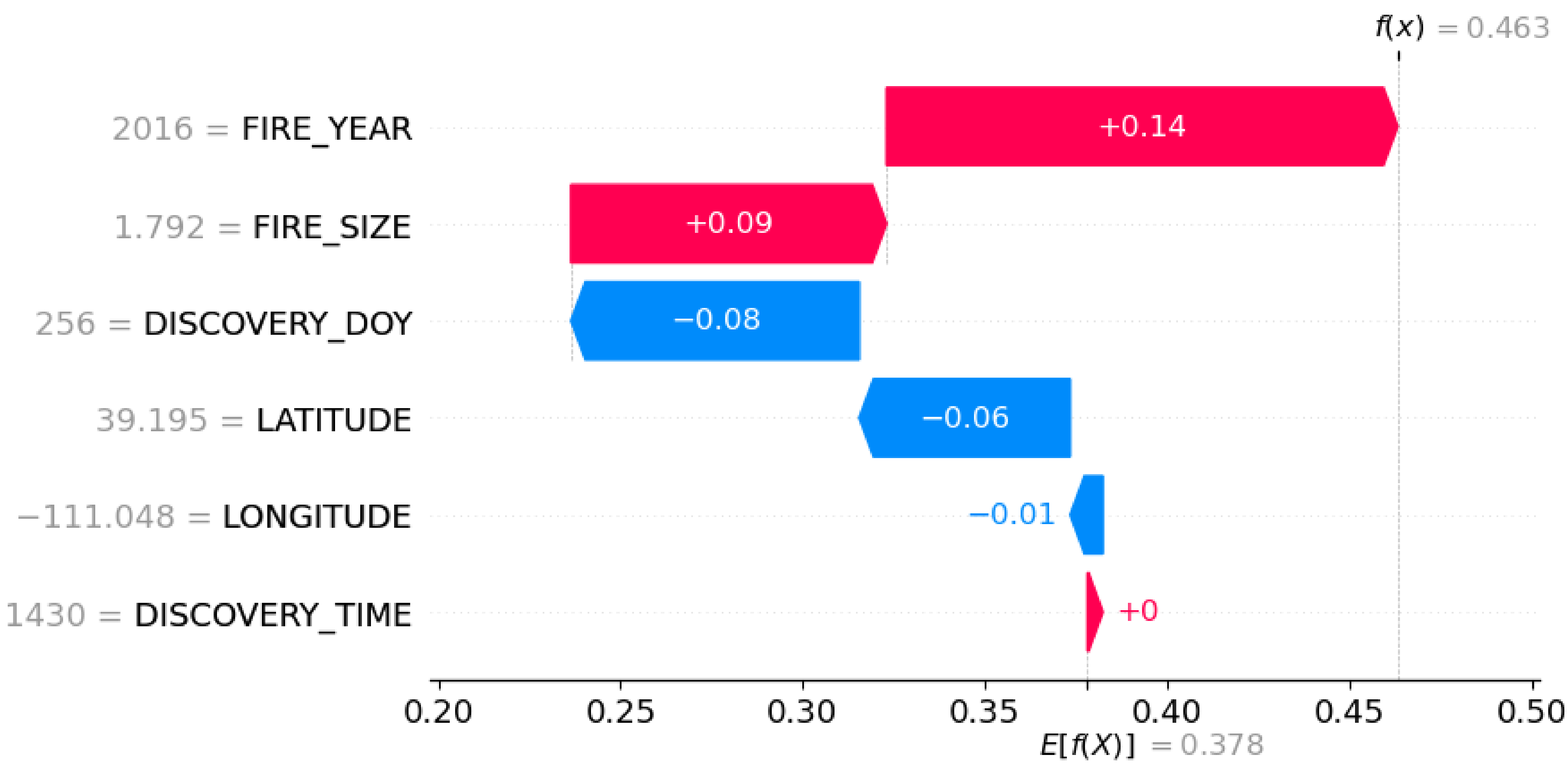
Global Importances:

Beeswarm Plot



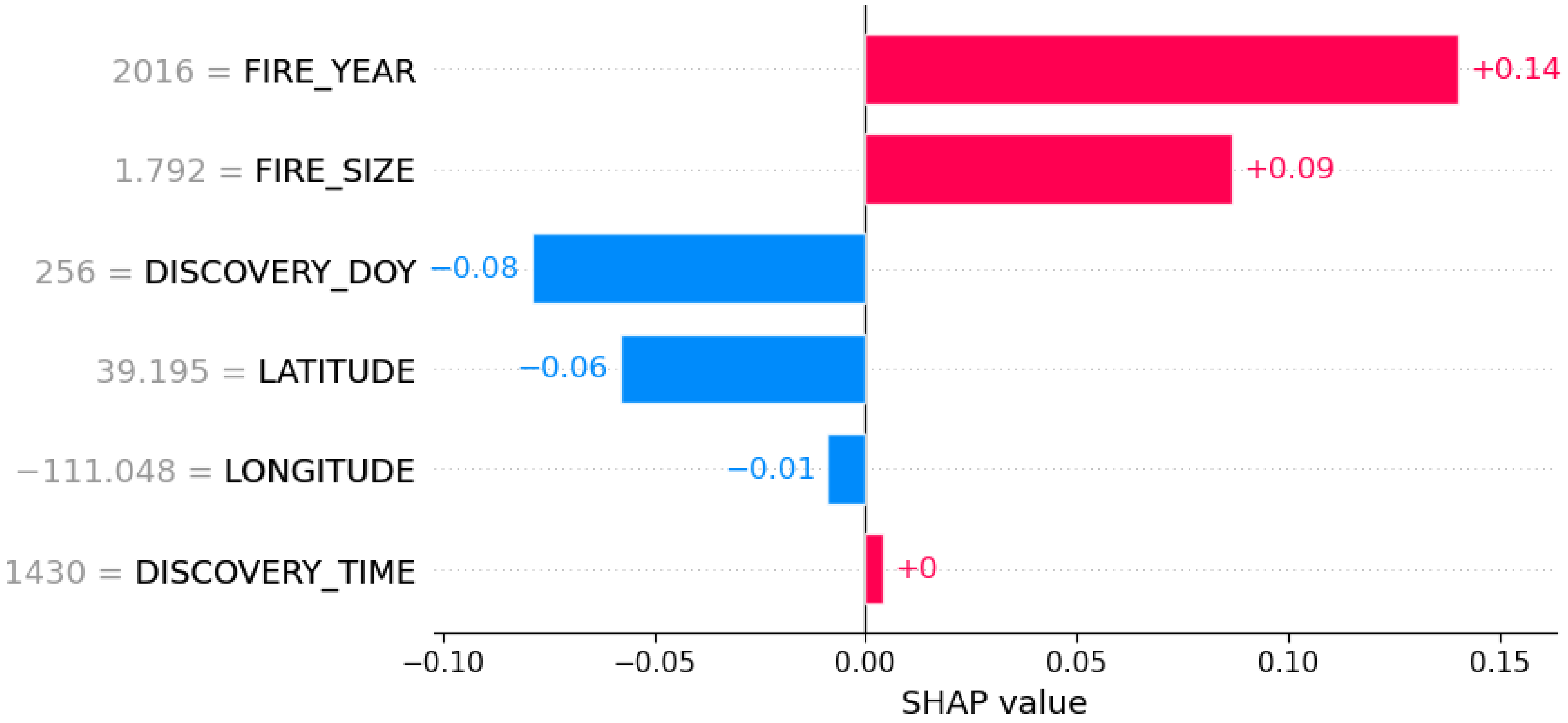
Local Importance: Waterfall plot

(Single prediction)



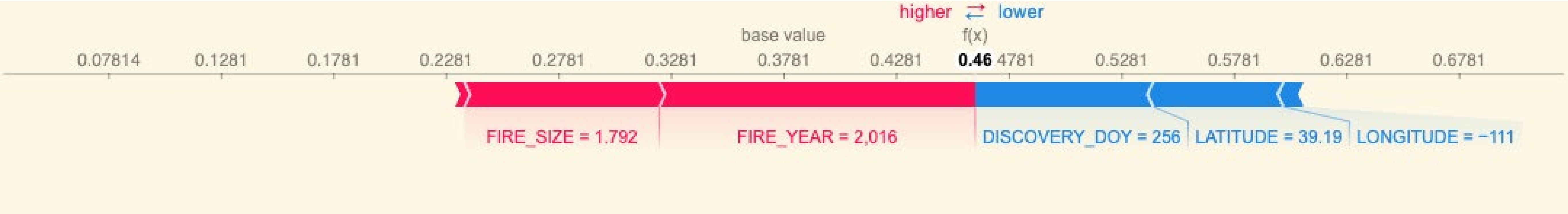
Local Importance: Bar plot

(Single prediction)



Local Importance: Force plot

(Single prediction)



SHAP Value Overview

SHAP gives us two views of a model:

- Global: which features matter overall
  - Bar plots tell us how much a feature matters.
  - Scatter and beeswarm plots tell us how it affects predictions
- Local: why a specific prediction happened
  - Waterfall plots show the model’s reasoning step by step.”

Disadvantages

- But it still has a few limitations:
  - Slow (especially kernel SHAP)
  - Can be hard to interpret
  - Misleading when features are correlated
    - SHAP assumes that features are independent
    - If features are correlated, SHAP can arbitrarily split their importance or attribute too much importance to one and downplaying the other

Summary

- SHAP is one way to help explain model predictions
- It is a powerful and theoretically sound method for interpreting machine learning models
- The shap library in python provides (fairly) fast implementation and lots of examples
- Be cautious of assigning causation